

Technology Stack

Date	15 March 2026 26 June 2026
Team ID	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Project Name	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, Jinja2 templates, CSS, JS CSS, JS <i>[Specify technology]</i>	No build step required; templates render server-side from keeping the stack lightweight for a fast demo turnaround. keeping the stack lightweight for a fast demo turnaround. <i>[Explain why this was chosen (e.g., responsive design, component reusability)]</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.10, FastAPI, Uvicorn Uvicorn <i>[Specify technology]</i>	Async-first framework with automatic OpenAPI docs, native Pydantic validation, and fast iteration for REST endpoints. Pydantic validation, and fast iteration for REST endpoints. <i>[Explain why this was chosen (e.g., fast processing, existing team expertise)]</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	JSON flat files (data/*.json) (data/*.json) <i>[Specify technology]</i>	Zero-setup persistence that mirrors a relational schema (Users, Queries, Responses, Quizzes, Summaries, Learning Paths) Queries, Responses, Quizzes, Summaries, Learning Paths) <i>[Explain why this was chosen without requiring an external DBMS for a demo/academic project. without requiring an external DBMS for a demo/academic project. (e.g., handles unstructured data, strong relational mapping)]</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Docker (python:3.10-slim) <i>[Specify technology]</i>	Containerization ensures the app runs consistently across development and evaluation environments; exposed on p development and evaluation environments; exposed on p <i>[Explain why this was chosen (e.g., easy scalability, cost-effective hosting)]</i>
5	Version Control & CI/CD	<i>[Specify technology]</i>	Standard collaborative source control for the team; no au CI pipeline is configured yet. CI pipeline is configured yet. <i>[Explain why this was chosen (e.g., seamless collaboration, automated testing)]</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API & local LaMini-Flan-T5 local LaMini-Flan-T5 <i>[Specify model technology]</i>	Combines cloud-grade reasoning quality (Gemini, with dy best-model resolution) with an offline-capable local mode best-model resolution) with an offline-capable local mode <i>[Explain the specific feature resilience. this enables in your project]</i>